Forecasting Average Traffic Speed on the PEMS08 Sensor Network

Near-future and long-term traffic speed forecasting

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Artificial Intelligence

Traffic Speed Forecasting as Graph Time Series

Task

Recent sensor history → future sensor speed?

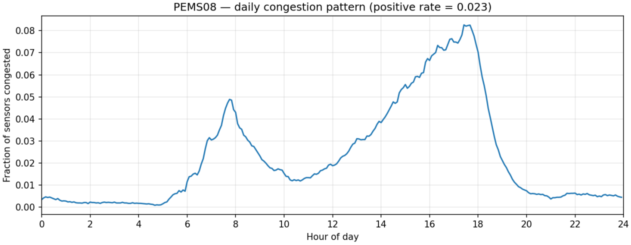
Forecast ranges

- near future: +5 to +60 min
- longer term: +1h, +3h, +6h, +12h

Graph view

- node-level regression
- sensor time series
- neighbour traffic context

Use case: travel time, congestion warnings, routing



Daily congestion pattern; positive rate: 2.27%.

PEMS08: Sparse, Connected Road Graph

PEMS08 benchmark

sensors	**170**
time steps	**17,856**
resolution	**5 min**
duration	**62 days**

Road graph facts

- 277 unique directed edges
- 1 weakly connected component
- avg in/out-degree: 1.63
- 16 Louvain communities

Modeling consequence

Sensor history + nearby-road context

NN1: 60-Minute Speed Forecast

Task

- one speed value per sensor
- horizon: +60 min
- input window: previous 12 readings = 1 hour

Features per sensor and time step

- flow, occupancy, speed
- time-of-day and day-of-week
- incoming-neighbour averages

Input/output shape

$$[12, 170, 10] \rightarrow [170]$$

NN1/NN2: Split and Leakage Control

Chronological split

- train / validation / test = 60% / 20% / 20%
- train = past, test = future
- no random k-fold on autocorrelated time series

Normalisation

- per-sensor z-score
- training statistics only
- metrics after denormalising to mph

Boundary gap

$$W = 12 \text{ steps}$$

Input windows never cross split boundaries.

NN1/NN2: GCN-GRU Architecture

Spatial step

$$H = \mathrm{ReLU}(AXW_{\mathrm{gcn}}), \qquad A = \alpha A_{\mathrm{fixed}} + (1-\alpha)\tilde{A}, \qquad \tilde{A} = \mathrm{softmax}(\mathrm{ReLU}(E_1 E_2^\top)).$$

Architecture

- 2 graph-convolution layers
- adaptive adjacency
- residual node signal
- 2-layer GRU over 12 steps

Training

- Huber loss, $\delta = 1$
- Adam, lr 10^{-3}
- early stopping on validation MAE
- seed 42

NN2: Full +5 to +60 Minute Curve

Task

- speed at 5, 10, . . . , 60 min
- one value per sensor and horizon
- same input window and features as NN1
- standard multi-horizon benchmark setup

Input/output shape

$$[12, 170, 10] \to [170, 12]$$

`-single-horizon` selects NN1; default mode trains NN2.

NN2: One Head Predicts All Horizons

Same core model

- spatial GCN
- adaptive adjacency + residual
- temporal GRU
- same split and normalisation

Only the output head changes

$$\text{Linear}(\mathrm{hidden} \to 1) \quad \Longrightarrow \quad \text{Linear}(\mathrm{hidden} \to 12)$$

Best run

- NN1: 58,954 parameters
- NN2: 59,669 parameters
- extra output horizons with almost no parameter cost

NN3: Later-Today Speed Forecasting

Goal: from immediate traffic control to later-today planning

Prediction horizons

+1h	**+3h**	**+6h**	**+12h**
12 steps	36 steps	72 steps	144 steps

Target

- speed per sensor
- 4 horizons jointly

Leakage-safe inputs

- flow, occupancy, speed at t
- speed lags: 3h, 6h, 12h, 1d, 2d, 7d
- current + target calendar time

Built dataset

X	$(15696, 170, 29)$
y	$(15696, 170, 4)$
train / val / test	9417 / 3139 / 3140

No future traffic measurements as inputs

NN3: Shared-Sensor GRU

2 hours of features $\rightarrow$ shared GRU $\rightarrow$ +1h, +3h, +6h, +12h speeds

Why this architecture

- small, stable model
- temporal patterns shared across sensors
- all long horizons in one pass

Best run

- input window: 24 steps = 2 hours
- hidden size: 64
- GRU layers: 2
- dropout: 0.2
- parameters: 43,460
- loss: Smooth L1 / Huber

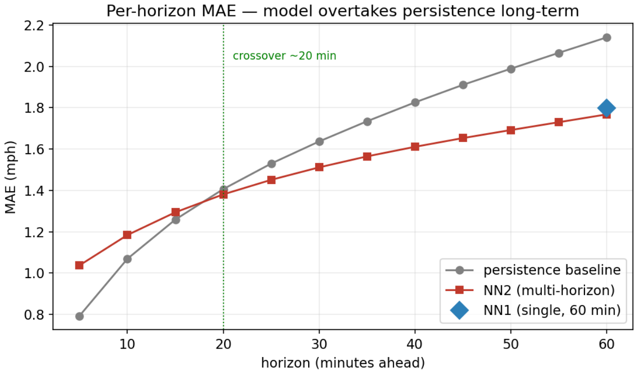
Train-only normalisation; metrics in mph

Results: NN1 vs NN2 vs Persistence

Model (test set)	MAE	RMSE	MAPE
Persistence (avg over 5–60 min)	1.6133	3.7811	3.08%
Persistence (60 min only)	2.1403	4.9692	4.21%
NN1 – single-horizon (60 min)	1.7979	4.4583	4.13%
NN2 – multi-horizon (60 min)	**1.7677**	4.2927	3.98%
NN2 – multi-horizon (avg)	**1.4897**	3.6037	3.23%

- 60 min: NN2 1.7677 vs. NN1 1.7979 MAE.
- 60-min persistence improvement: 17.4%.
- average persistence improvement: 7.7%.

Results: Per-Horizon Crossover



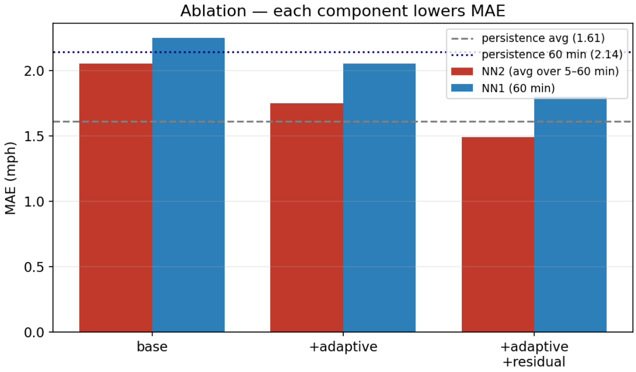
Pattern

- 5–15 min: persistence wins
- $\sim$20 min: crossover
- 60 min: NN2 wins by 0.3726 MAE

Reason

Short-term speed changes slowly; longer horizons need learned temporal-spatial patterns.

Results: Architecture Ablation



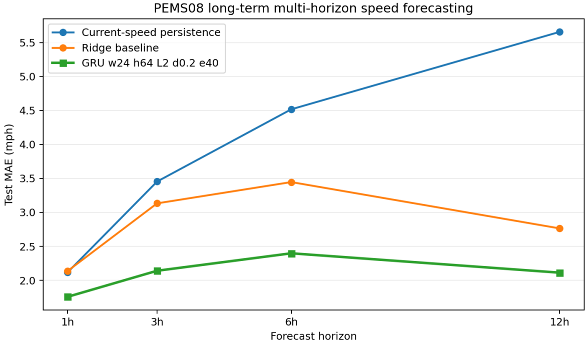
Pattern

- fixed graph: weakest
- + adaptive: lower MAE
- + residual: lower MAE again

Takeaway

Architecture gains, not only model scale.

Results: NN3 Long-Horizon Forecasts



Metric	Ridge	GRU
Avg MAE	2.8699	2.1015
+1h MAE	2.1364	1.7566
+3h MAE	3.1328	2.1402
+6h MAE	3.4462	2.3974
+12h MAE	2.7640	2.1120

Takeaways

- PEMS08: graph time-series forecasting
- short horizons: immediate traffic dynamics
- NN3: later-today horizons (+1h, +3h, +6h, +12h)
- chronological splits + leakage checks
- best long-horizon GRU: lowest MAE at every horizon